Algoritmekader Data Exploration

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1 Algoritmekader Data Preparation

1.1 Purpose and approach

The Algoritmekader source is a list of 63 Markdown files with YAML front matter. The website is designed to read one requirement at a time. This research views it as a system. Thus this data preparation takes the raw, machine-readable dataset and aims to format it so global inspection is possible. The Markdown formatting used by the Algoritmekader represents a semi-structured data format, with some fields strictly defined and `Toelichting` containing free text. The goal was to create a structured dataset for all the *vereisten* listed in the Algoritmekader while retaining provenance and all information contained within the source material.

Data preparation followed an iterative, exploratory process. Accordingly, the initial extraction was unfiltered. This allowed for a complete inspection of the data, which was not possible through the Algoritmekader website. Subsequent iterations introduced transformations after inspection of the resulting dataset and comparison with the Algoritmekader website.

Iterative processing was required because:

- no schema or overview of the data structure was available; only the source files and the Algoritmekader website architecture were available;
- website-generation metadata and semantic content coexisted in the Markdown source material;
- not all extracted fields represented the same kind of information.

New versions were created using the following general process. The steps were iterative rather than strictly chronological:

1. extract information directly from the Algoritmekader Git repository;
2. clean and transform the data in Python;
3. export the resulting data to CSV and Excel;

4. inspect the exported data manually for inconsistencies and source-specific issues;

5. process the CSV using a Python script to generate LaTeX tables automatically for inspection and documentation;

6. interpret the generated tables and use the findings to inform the next version.

1.2 Staging Summary

Table 1: Staged transformation of the raw Algoritmekader source into the ontology specification.

Stage	**Purpose**	**Processing and decisions**	**Result carried forward**
Raw Algoritmekader	Establish the complete source material before imposing an ontology structure.	The source consists of Markdown requirement files with YAML front matter, Markdown sections, website-generation directives, controlled values, free text, and links. The website presents requirements individually and does not provide a global schema for the collection.	Authoritative source material for exploratory extraction.
V1: exploratory extraction	Determine what information is present and how consistently the source is structured.	All 63 Markdown files were parsed without filtering. YAML keys became columns, every `##` heading became a section field, links were extracted, and nested YAML values were serialized for CSV/Excel inspection. Column profiles, low-cardinality values, metadata/section inventories, and a preliminary cardinality-based field classification were generated.	A complete but deliberately unrefined tabular representation exposing source structure, inconsistencies, and extraction artefacts.

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Table 1 continued

Stage	Purpose	Processing and decisions	Result carried forward
V2: structural normalization	Correct extraction artefacts while preserving the meaning and provenance of the 62 actual requirements.	`index.md` is excluded; identity is normalized to `requirement_id`, `title`, and `source_file`; website-only metadata is removed; explanatory one-off headings are merged into `section_toelichting`; list-valued metadata is preserved and exploded into individual assignments; and field types are classified explicitly from source inspection. `section_maatregelen`, `section_hulpmiddelen`, `uitzondering`, and `status_vereiste` are retained for later semantic treatment.	A source-faithful, structurally normalized requirement dataset and explicit vocabulary assignments suitable for semantic analysis.
V3: semantic normalization and source expansion *(planned)*	Resolve structures that V2 showed cannot be represented adequately as ordinary requirement attributes.	Import the separate maatregelen and hulpmiddelen collections and resolve their generated references; import the canonical `begrippenlijst.md`; distinguish substantive applicability text from generation directives in `section_van toepassing op`; inspect sparse status metadata; and formalize compound applicability and exception expressions as conditions rather than categorical properties.	Normalized entities, terminology, relations, and applicability structures from the Algoritmekader source family.
Ontology modelling and alignment *(planned)*	Translate the stabilized source model into explicit ontology constructs and determine what can be reused from existing vocabularies.	Define candidate classes, properties, controlled concepts, provenance relations, and applicability structures from the normalized data. Compare these concepts with relevant external vocabularies, including the DPV EU AI Act vocabulary, and reuse or specialize existing terms where their semantics match the required concepts.	A candidate ontology model with documented reuse, extensions, mappings, and unresolved modelling decisions.
Final ontology specification *(planned)*	Produce the formal specification against which the ontology implementation can be evaluated.	Consolidate the validated conceptual model into explicit definitions of classes, properties, relations, vocabularies, applicability rules, provenance, identifiers, and external mappings. Document modelling decisions and constraints required for consistent implementation and extension.	The final ontology specification forming the basis for implementation, validation, and later audit-support applications.

2 Version 1: Initial Scraping

2.1 Objective

Version 1 was an exploratory extraction. The questions guiding this version were:

- What fields existed in the YAML front matter?
- What Markdown sections occurred?
- How consistently did these fields and Markdown sections occur?
- Which fields appeared to be categorical?
- Could the source structure be translated straightforwardly into ontology concepts and relations?

2.2 Procedure

The first iteration of the `mine_vereisten` Python script did the following:

- collected all 63 `.md` files in the `vereisten` directory of the Algoritmekader Git repository;
- separated YAML front matter from the Markdown body;
- extracted YAML keys into columns;
- interpreted each `##` heading as a separate section;
- extracted Markdown links;
- serialized nested YAML values for CSV and Excel export;
- generated five descriptive tables:
  - **Column profile:** identify columns requiring additional processing or deletion;
  - **Observed low-cardinality values:** inspect the values contained in categorical or potentially categorical fields;
  - **Preliminary field classification:** inspect the apparent type of information contained in each column;
  - **Metadata fields:** identify fields originating from YAML front matter;
  - **Section fields:** identify the Markdown sections occurring across source files.

2.3 Observations

1. **Four identity/provenance fields.** The fields `filename`, `slug`, `title`, and `id` all contain the same information. None of the fields contain only the requirement ID. V2 should extract the tagged requirement for a sole identity field and maintain one field to register provenance from the Algoritmekader.

2. **Scope issue.** The file `index.md` was parsed as if it were a requirement, although it is an index page. This introduced one non-requirement record and several index-specific fields. V2 therefore excludes `index.md` before parsing.

3. **Website metadata.** The fields `hide`, `allow_excel_export`, and `summary` describe website configuration or presentation rather than requirement content. The field `open-source` contained the same three categorical values for all 62 requirement records and therefore provided no discriminatory information in the extracted dataset. These fields were removed in V2.

4. **Markdown headings were over-parsed.** Every `##` heading was initially converted into a separate field. In `aia-38`, this created `section_testen_van_ai_systemen_met_een_hoog_risico_onder_reÃńle_omstandigheden_buiten_ai_testomgevingen`; in `avg-10`, it created `section_algemene_eisen_aan_geautomatiseerde_besluitvorming_door_overheden`. Inspection of the Algoritmekader website showed that both were subsections of the explanatory text. V2 therefore merges their contents into `section_toelichting`.

5. **Multi-valued categorical fields were serialized as single values.** This affected `levenscyclus`, `onderwerp`, `soort-toepassing`, `risicogroep`, `rol-ai-act`, `transparantieverplichting`, `systeemrisico`, and `rollen`. For example, `["governance", "transparantie"]` was counted as one value rather than two assignments. Different orderings, such as `["governance", "transparantie"]` and `["transparantie", "governance"]`, were also treated as distinct. V2 preserves these fields as lists and expands them into individual requirement–field–value assignments before analysis.

6. **Field classification based on cardinality was unreliable.** V1 classified fields using the heuristic ≤ 2 unique values = Boolean, ≤ 30 = categorical, otherwise = free text. This misclassified fields such as `levenscyclus`, whose serialized list combinations produced 32 unique values, and `uitzondering`, whose low cardinality masked rule-like content. V2 therefore uses an explicit semantic classification informed by source inspection.

7. **Generated relationship sections were mistaken for content fields.** The fields `section_maatregelen` and `section_hulpmiddelen` mainly contained website-generation directives rather than substantive prose. They therefore indicate links to separate source entities rather than literal requirement content. Future iterations should import the corresponding

`maatregelen` and `hulpmiddelen` sources separately and represent these as related entities.

2.4 Tables

2.4.1 Column Profile

Table 2: Profile of fields extracted from Algoritmekader requirements.

Column	Non-empty	Total rows	Unique values	Coverage (%)
filename	63	63	63	100.0
slug	63	63	63	100.0
title	63	63	63	100.0
id	63	63	63	100.0
toelichting	62	63	62	98.4
levenscyclus	62	63	32	98.4
onderwerp	62	63	25	98.4
soort-toepassing	62	63	4	98.4
risicogroep	62	63	8	98.4
rol-ai-act	62	63	3	98.4
transparantieverplichting	62	63	4	98.4
systeemrisico	62	63	4	98.4
open-source	62	63	1	98.4
hide	62	63	2	98.4
rollen	62	63	15	98.4
section_vereiste	62	63	62	98.4
section_toelichting	62	63	62	98.4
section_bronnen	62	63	62	98.4
section_van toepassing op	61	63	5	96.8
section_risico	62	63	56	98.4
section_maatregelen	62	63	62	98.4
uitzondering	38	63	2	60.3
section_voorbeeld	2	63	2	3.2
section_hulpmiddelen	5	63	5	7.9
section_testen van ai-systemen met een hoog risico onder reële omstandigheden buiten ai-testomgevingen	1	63	1	1.6
status_vereiste	2	63	1	3.2
section_algemene eisen aan geautomatiseerde besluitvorming door overheden	1	63	1	1.6
section_begrippen	1	63	1	1.6
summary	1	63	1	1.6
allow_excel_export	1	63	1	1.6
section_wetten en regels	1	63	1	1.6
section_aantal vereisten verschilt per situatie	1	63	1	1.6

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Table 2 continued

Column	Non-empty	Total rows	Unique values	Coverage (%)
section_help ons deze pagina te verbeteren	1	63	1	1.6

2.4.2 Controlled Values

Table 3: Observed values for low-cardinality metadata fields in the Algoritmekader requirements.

Field	Value	Count
onderwerp	["transparantie"]	11
onderwerp	["technische-robuustheid-en-veiligheid"]	9
onderwerp	["governance"]	7
onderwerp	["privacy-en-gegevensbescherming"]	6
onderwerp	["data"]	3
onderwerp	["fundamentele-rechten"]	3
onderwerp	["menselijke-controle", "technische-robuustheid-en-veiligheid"]	2
onderwerp	["menselijke-controle"]	2
onderwerp	["governance", "transparantie"]	2
onderwerp	["governance", "menselijke-controle"]	2
onderwerp	["transparantie", "technische-robuustheid-en-veiligheid"]	1
onderwerp	["menselijke-controle", "governance"]	1
onderwerp	["fundamentele-rechten", "bias-en-non-discriminatie"]	1
onderwerp	["governance", "fundamentele-rechten", "transparantie"]	1
onderwerp	["transparantie", "governance"]	1
onderwerp	["transparantie", "data"]	1
onderwerp	["fundamentele-rechten", "privacy-en-gegevensbescherming"]	1
onderwerp	["governance", "privacy-en-gegevensbescherming"]	1
onderwerp	["privacy-en-gegevensbescherming", "transparantie"]	1
onderwerp	["privacy-en-gegevensbescherming", "bias-en-non-discriminatie"]	1
onderwerp	["privacy-en-gegevensbescherming", "data"]	1
onderwerp	["privacy-en-gegevensbescherming", "menselijke-controle"]	1
onderwerp	["privacy-en-gegevensbescherming", "technische-robuustheid-en-veiligheid"]	1
onderwerp	["fundamentele-rechten", "menselijke-controle"]	1
onderwerp	["bias-en-non-discriminatie"]	1
soort-toepassing	["ai-systeem", "ai-systeem-voor-algemene-doeleinden"]	30
soort-toepassing	["ai-model-voor-algemene-doeleinden", "ai-systeem", "ai-systeem-voor-algemene-doeleinden"]	23

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Table 3 continued

Field	Value	Count
soort-toepassing	["ai-model-voor-algemene-doeleinden"]	5
soort-toepassing	["ai-systeem", "ai-systeem-voor-algemene-doeleinden", "ai-model-voor-algemene-doeleinden"]	4
risicogroep	["hoog-risico-ai-systeem"]	28
risicogroep	["hoog-risico-ai-systeem", "geen-hoog-risico-ai-systeem", "uitzondering-van-toepassing", "verboden-ai"]	23
risicogroep	["niet-van-toepassing"]	5
risicogroep	["hoog-risico-ai-systeem", "geen-hoog-risico-ai-systeem", "niet-van-toepassing"]	2
risicogroep	["geen-hoog-risico-ai-systeem"]	1
risicogroep	["hoog-risico-ai-systeem", "verboden-ai", "geen-hoog-risico-ai-systeem", "niet-van-toepassing"]	1
risicogroep	["verboden-ai"]	1
risicogroep	["hoog-risico-ai-systeem", "geen-hoog-risico-ai-systeem"]	1
rol-ai-act	["aanbieder"]	28
rol-ai-act	["aanbieder", "gebruiksverantwoordelijke", "importeur", "distributeur"]	26
rol-ai-act	["gebruiksverantwoordelijke"]	8
transparantieverplichting	["geen-transparantieverplichting", "transparantieverplichting"]	30
transparantieverplichting	["geen-transparantieverplichting", "transparantieverplichting", "niet-van-toepassing"]	26
transparantieverplichting	["niet-van-toepassing"]	5
transparantieverplichting	["transparantieverplichting"]	1
systeemrisico	["niet-van-toepassing"]	30
systeemrisico	["systeemrisico", "geen-systeemrisico", "niet-van-toepassing"]	27
systeemrisico	["systeemrisico"]	3
systeemrisico	["systeemrisico", "geen-systeemrisico"]	2
open-source	["open-source", "geen-open-source", "niet-van-toepassing"]	62
hide	["navigation"]	61
hide	["navigation", "toc"]	1
rollen	["projectleider"]	25
rollen	["projectleider", "ontwikkelaar"]	8
rollen	["projectleider", "beleid-en-advies"]	6
rollen	["jurist", "projectleider"]	4
rollen	["ontwikkelaar", "projectleider"]	3
rollen	["jurist", "ontwikkelaar"]	3
rollen	["jurist"]	3
rollen	["projectleider", "jurist"]	2
rollen	["ontwikkelaar"]	2
rollen	["projectleider", "beleid-en-advies", "ontwikkelaar"]	1
rollen	["projectleider", "ontwikkelaar", "beleid-en-advies"]	1
rollen	["ontwikkelaar", "beleid-en-advies"]	1

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Table 3 continued

Field	Value	Count
rollen	["projectleider", "jurist", "beleid-en-advies"]	1
rollen	["beleid-en-advies", "projectleider", "jurist", "ontwikkelaar"]	1
rollen	["beleid-en-advies", "ontwikkelaar"]	1
uitzondering	["risicogroep-uitzondering-van-toepassing"]	33
uitzondering	["risicogroep-uitzondering-van-toepassing", "(\"soort-toepassing-ai-systeem-voor-algemene-doeleinden\" || \"soort-toepassing-ai-systeem\") && \"open-source-open-source\" && \"transparantieverplichting-geen-transparantieverplichting\" && \"risicogroep-geen-hoog-risico-ai-systeem\""]	5
status_vereiste	["geldend"]	2
summary	Overzicht van vereisten voor overheden die algoritmes of AI ontwikkelen of gebruiken.	1
allow_excel_export	True	1

2.4.3 Field Classification

Table 4: Initial structural classification of fields extracted from the Algoritmekader.

Field	Category	Non-empty	Unique values
filename	Identity / provenance	63	63
slug	Identity / provenance	63	63
title	Identity / provenance	63	63
id	Identity / provenance	63	63
toelichting	Free text / identifier	62	62
levenscyclus	Free text / identifier	62	32
onderwerp	Controlled / categorical metadata	62	25
soort-toepassing	Controlled / categorical metadata	62	4
risicogroep	Controlled / categorical metadata	62	8
rol-ai-act	Controlled / categorical metadata	62	3
transparantieverplichting	Controlled / categorical metadata	62	4
systeemrisico	Controlled / categorical metadata	62	4
open-source	Boolean / binary metadata	62	1
hide	Boolean / binary metadata	62	2
rollen	Controlled / categorical metadata	62	15
section_vereiste	Markdown section	62	62
section_toelichting	Markdown section	62	62
section_bronnen	Markdown section	62	62

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Table 4 continued

Field	Category	Non-empty	Unique values
section_van toepassing op	Markdown section	61	5
section_risico	Markdown section	62	56
section_maatregelen	Markdown section	62	62
uitzondering	Boolean / binary metadata	38	2
section_voorbeeld	Markdown section	2	2
section_hulpmiddelen	Markdown section	5	5
section_testen van ai-systemen met een hoog risico onder reële omstandigheden buiten ai-testomgevingen	Markdown section	1	1
status_vereiste	Boolean / binary metadata	2	1
section_algemene eisen aan geautomatiseerde besluitvorming door overheden	Markdown section	1	1
section_begrippen	Markdown section	1	1
summary	Boolean / binary metadata	1	1
allow_excel_export	Boolean / binary metadata	1	1
section_wetten en regels	Markdown section	1	1
section_aantal vereisten verschilt per situatie	Markdown section	1	1
section_help ons deze pagina te verbeteren	Markdown section	1	1

2.4.4 Metadata Fields

Table 5: Metadata fields present in Algoritmekader requirement files.

Metadata field
filename
slug
title
id
toelichting
levenscyclus
onderwerp
soort-toepassing
risicogroep
rol-ai-act
transparantieverplichting
systeemrisico
open-source
hide
rollen
uitzondering

Table 5 continued

Metadata field
status_vereiste
summary
allow_excel_export

2.4.5 Section Fields

Table 6: Markdown sections identified in Algoritmekader requirement files.

Markdown section
section_vereiste
section_toelichting
section_bronnen
section_van toepassing op
section_risico
section_maatregelen
section_voorbeeld
section_hulpmiddelen
section_testen van ai-systemen met een hoog risico onder reële omstandigheden buiten ai-testomgevingen
section_algemene eisen aan geautomatiseerde besluitvorming door overheden
section_begrippen
section_wetten en regels
section_aantal vereisten verschilt per situatie
section_help ons deze pagina te verbeteren

3 Version 2: Data Clean-up First Pass

3.1 Tables

3.1.1 Column Profile

Table 7: Profile of fields extracted from Algoritmekader requirements.

Column	Non-empty	Total rows	Unique values	Coverage (%)
requirement_id	62	62	62	100.0
title	62	62	62	100.0
source_file	62	62	62	100.0

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Table 7: Profile of fields extracted from Algoritmekader requirements.

Column	Non-empty	Total rows	Unique values	Coverage (%)
toelichting	62	62	62	100.0
levenscyclus	62	62	9	100.0
onderwerp	62	62	8	100.0
soort-toepassing	62	62	3	100.0
risicogroep	62	62	5	100.0
rol-ai-act	62	62	4	100.0
transparantieverplichting	62	62	3	100.0
systeemrisico	62	62	3	100.0
rollen	62	62	4	100.0
section_vereiste	62	62	62	100.0
section_toelichting	62	62	62	100.0
section_bronnen	62	62	62	100.0
section_van toepassing op	61	62	5	98.4
section_risico	62	62	56	100.0
section_maatregelen	62	62	62	100.0
uitzondering	38	62	2	61.3
section_hulpmiddelen	5	62	5	8.1
status_vereiste	2	62	1	3.2

3.1.2 Controlled Values

Table 8: Individual controlled vocabulary values observed in the Algoritmekader requirements.

Field	Value	Count	Coverage (%)
levenscyclus	ontwerp	28	45.2
levenscyclus	monitoring-en-beheer	27	43.6
levenscyclus	ontwikkelen	16	25.8
levenscyclus	organisatieverantwoordelijkheden	15	24.2
levenscyclus	dataverkenning-en-datapreparatie	13	21.0
levenscyclus	implementatie	12	19.4
levenscyclus	verificatie-en-validatie	11	17.7
levenscyclus	uitfaseren	4	6.4
levenscyclus	probleemanalyse	2	3.2
onderwerp	transparantie	18	29.0
onderwerp	governance	15	24.2
onderwerp	privacy-en-gegevensbescherming	13	21.0
onderwerp	technische-robuustheid-en-veiligheid	13	21.0
onderwerp	menselijke-controle	9	14.5
onderwerp	fundamentele-rechten	7	11.3
onderwerp	data	5	8.1
onderwerp	bias-en-non-discriminatie	3	4.8
risicogroep	hoog-risico-ai-systeem	55	88.7
risicogroep	geen-hoog-risico-ai-systeem	28	45.2

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Table 8: Individual controlled vocabulary values observed in the Algoritmekader requirements.

Field	Value	Count	Coverage (%)
risicogroep	verboden-ai	25	40.3
risicogroep	uitzondering-van-toepassing	23	37.1
risicogroep	niet-van-toepassing	8	12.9
rol-ai-act	aanbieder	54	87.1
rol-ai-act	gebruiksverantwoordelijke	34	54.8
rol-ai-act	distributeur	26	41.9
rol-ai-act	importeur	26	41.9
rollen	projectleider	52	83.9
rollen	ontwikkelaar	21	33.9
rollen	jurist	14	22.6
rollen	beleid-en-advies	12	19.4
soort-toepassing	ai-systeem	57	91.9
soort-toepassing	ai-systeem-voor-algemene-doeleinden	57	91.9
soort-toepassing	ai-model-voor-algemene-doeleinden	32	51.6
status_vereiste	geldend	2	3.2
systeemrisico	niet-van-toepassing	57	91.9
systeemrisico	systeemrisico	32	51.6
systeemrisico	geen-systeemrisico	29	46.8
transparantieverplichting	transparantieverplichting	57	91.9
transparantieverplichting	geen-transparantieverplichting	56	90.3
transparantieverplichting	niet-van-toepassing	31	50.0

3.1.3 Field Classification

Table 9: Structural classification of fields extracted from the Algoritmekader.

Field	Category	Non-empty	Unique values
requirement_id	Identity	62	62
title	Human-readable label	62	62
source_file	Source / provenance	62	62
toelichting	Descriptive text	62	62
levenscyclus	Controlled vocabulary	62	9
onderwerp	Controlled vocabulary	62	8
soort-toepassing	Applicability vocabulary	62	3
risicogroep	Applicability vocabulary	62	5
rol-ai-act	Applicability vocabulary	62	4
transparantieverplichting	Applicability vocabulary	62	3
systeemrisico	Applicability vocabulary	62	3
rollen	Organisational role vocabulary	62	4
section_vereiste	Normative statement	62	62
section_toelichting	Explanatory text	62	62
section_bronnen	Source / provenance	62	62
section_van toepassing op	Applicability note	61	5

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Table 9: Structural classification of fields extracted from the Algoritmekader.

Field	Category	Non-empty	Unique values
section_risico	Risk description	62	56
section_maatregelen	Relationship placeholder	62	62
uitzondering	Applicability rule	38	2
section_hulpmiddelen	Relationship placeholder	5	5
status_vereiste	Status metadata	2	1

3.1.4 Metadata Fields

Table 10: Metadata fields present in Algoritmekader requirement files.

Metadata field
requirement_id
title
source_file
toelichting
levenscyclus
onderwerp
soort-toepassing
risicogroep
rol-ai-act
transparantieverplichting
systeemrisico
rollen
uitzondering
status_vereiste

3.1.5 Section Fields

Table 11: Markdown sections identified in Algoritmekader requirement files.

Markdown section
section_vereiste
section_toelichting
section_bronnen
section_van toepassing op
section_risico
section_maatregelen
section_hulpmiddelen